



FATIMA JINNAH WOMEN UNIVERSITY RAWALPINDI

Department Of Software Engineering

Cloud Computing Lab

Saira Ejaz [2023-BSE-057]

Section: 5-B

LAB NO 8



AWS: Account Setup, IAM, VPC Inventory,
EC2, Docker & Gitea

Lab Tasks:

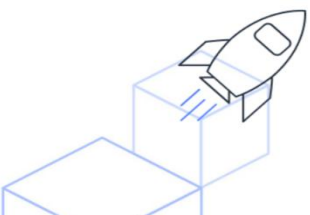
Task 1 — Create an AWS account and enable UAE (me-central-1)

1. Open your browser and go to: [AWS Signup](#)



Try AWS at no cost for up to 6 months

Start with USD \$100 in AWS credits, plus earn up to USD \$100 by completing various activities.



Sign up for AWS

Root user email address
Used for account recovery and as described in the [AWS Privacy Notice](#)

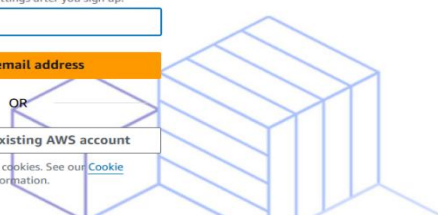
AWS account name
Choose a name for your account. You can change this name in your account settings after you sign up.

Verify email address

OR

Sign in to an existing AWS account

This site uses essential cookies. See our [Cookie Notice](#) for more information.



2. Complete registration (Account type: Personal, Plan: AWS Paid Plan), fill contact, billing (credit card) and phone details, complete verification. After successful registration capture:



Sign up for AWS

Choose your account plan

Free (6 months)	Paid
	
Free (6 months) Learn, experiment, and build prototypes	Paid Develop production-ready workloads
✓ Receive up to \$200 in credits	✓ Receive up to \$200 in credits
✓ Includes free usage of select services	✓ Includes free usage of select services
✗ Workloads scale beyond credit thresholds	✓ Workloads scale beyond credit thresholds
✗ Access to all AWS services and features	✓ Access to all AWS services and features
① After the 6 month free period or when all credits are used, you can choose to upgrade to a paid plan. Otherwise, your account closes automatically.	① After all of your credits are used, you are charged using <u>pay-as-you-go</u> pricing.



Why is this required?

Our verification process holds USD \$1 (or equivalent) for 3-5 days to verify your account and prevent fraud.

For the free plan, no charges occur until upgrade to a paid plan. Providing your billing information now enables a seamless upgrade to a paid plan.



Sign up for AWS

Billing Information

Billing country

Your billing country determines the payment methods available to you to pay for AWS services.

Pakistan ▼

Credit or Debit card number

ⓧ The credit card number is required.



AWS accepts most major credit and debit cards. To learn more about payment options, review our [FAQ](#)

Expiration date

March ▼ 2035 ▼

Security code ①

ⓧ A security code is required.

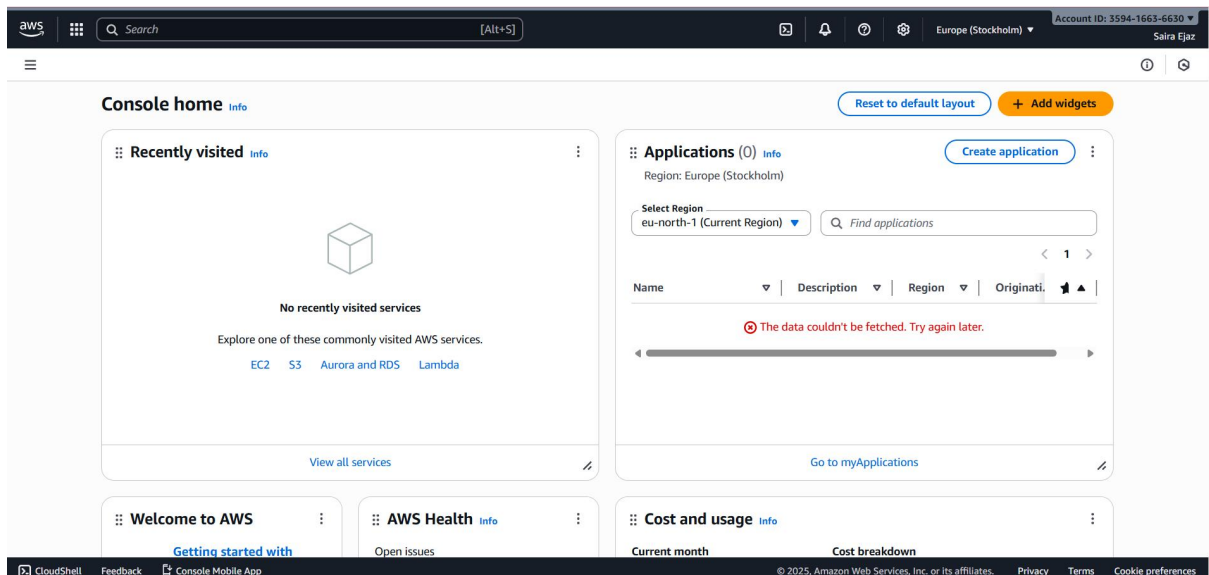
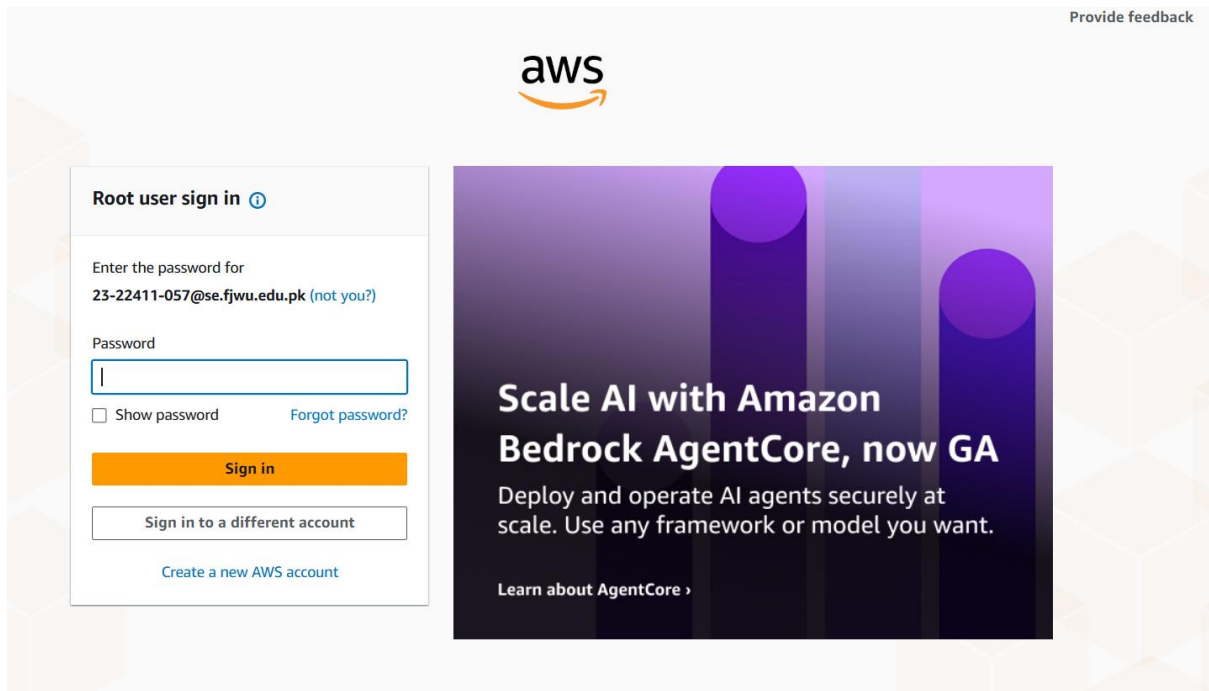
Cardholder's name

Saira Ejaz

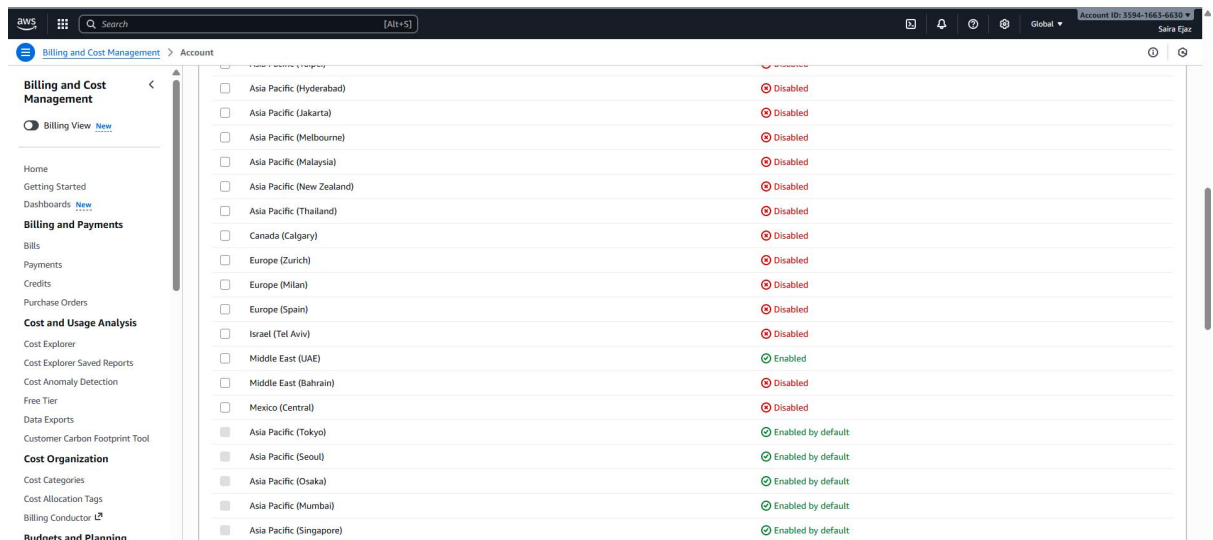
Billing address

☒ Use my contact address

3. Sign in as the root user (root email). Immediately capture:



4. From the Console, open the region selector and enable UAE (me-central-1), then switch to me-central-1. Capture the change:



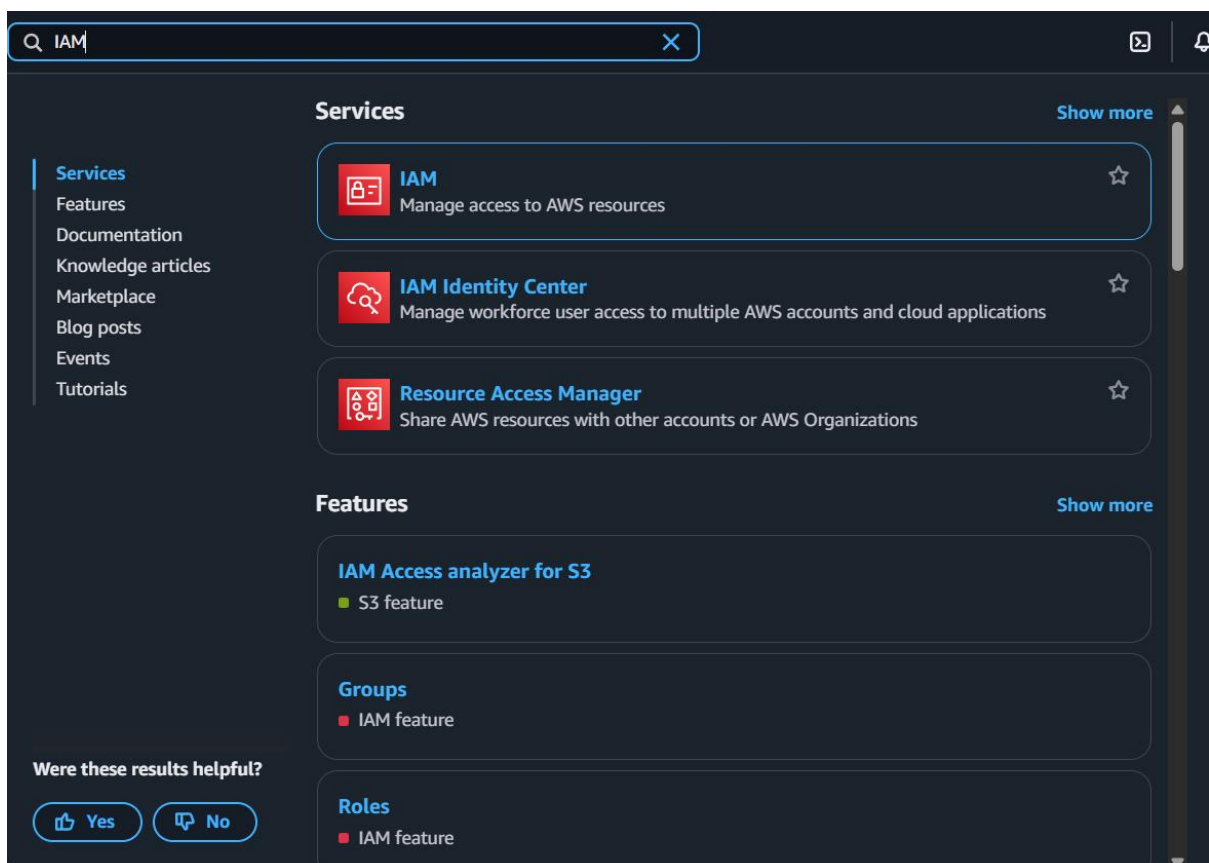
5. Task 1 summary screenshot (combine evidence):



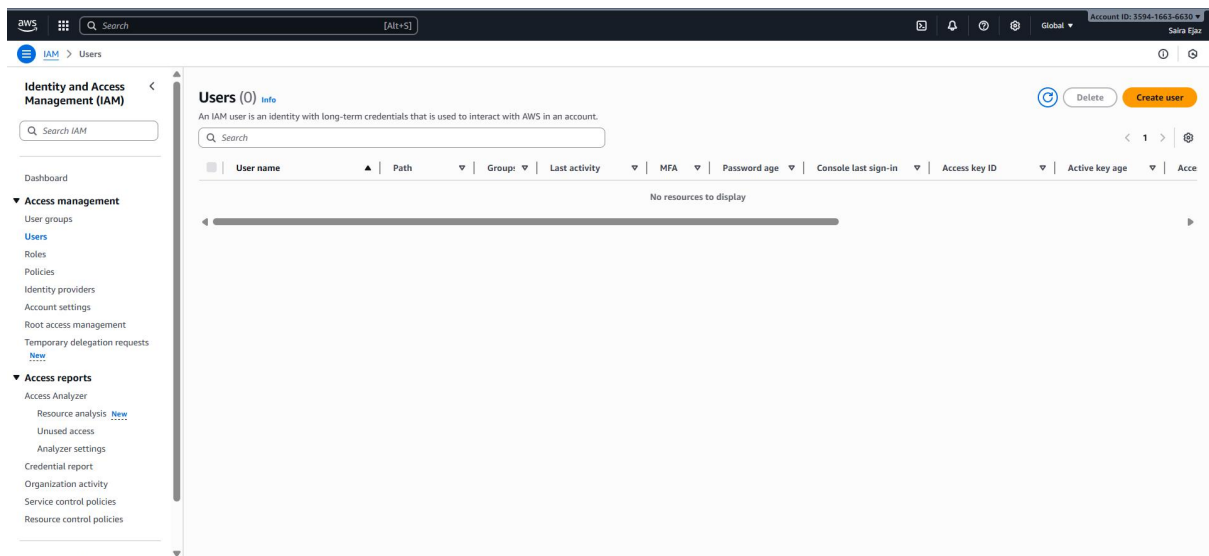
Task 2 — Create IAM Admin and Lab8User with console access

Goal: Create an IAM user named Admin (console access + Administrator Access policy), download the .csv, logout root, then login as the IAM Admin and create Lab8User with same permissions.

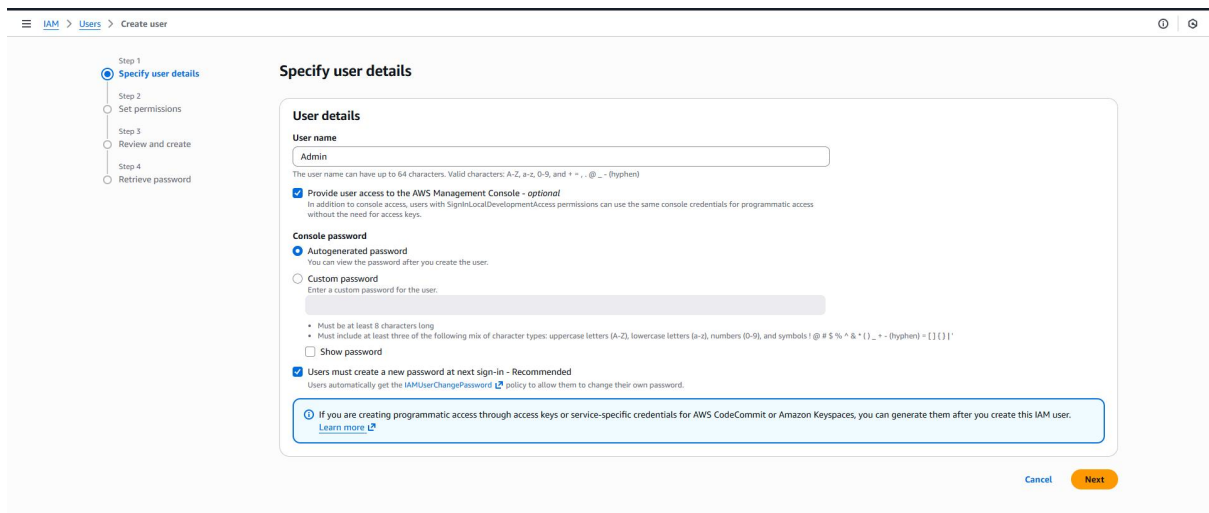
1. Open IAM via Console search (Alt+S → "IAM").



2. Create the Admin user: IAM → Users → Create user. Fill:



- Username: Admin
- Provide user access to the AWS Management Console
- Set console password (autogenerate or set)



- Attach policies directly → AdministratorAccess

Step 1

Specify user details

Step 2

Set permissions

Step 3

Review and create

Step 4

Retrieve password

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

☐ Add user to group

Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ Copy permissions

Copy all group memberships, attached managed policies, and inline policies from an existing user.

☒ Attach policies directly

Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (1/1423)

Create policy

Choose one or more policies to attach to your new user.

Search

Filter by Type

All types

< 1 2 3 4 5 6 7 ... 72 >

☒

AdministratorAccess

AWS managed - job function

0

☐

PowerUserAccess

AWS managed - job function

0

☐

ReadOnlyAccess

AWS managed - job function

0

☐

AWSCloudFormationReadOnlyAccess

AWS managed

0

☐

CloudFrontFullAccess

AWS managed

0

☐

AWSCloudHSMFullAccess

AWS managed

0

☐

AWSCloudHSMReadOnlyAccess

AWS managed

0

☐

ResourceGroupsandTagEditorFullAccess

AWS managed

0

Step 1

Specify user details

Step 2

Set permissions

Review and create

Step 4

Retrieve password

Review and create

Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.

User details

User name

Admin

Console password type

Autogenerated

Require password reset

Yes

Permissions summary

Name

Type

Used as

AdministratorAccess

AWS managed - job function

Permissions policy

IAMUserChangePassword

AWS managed

Permissions policy

Tags - optional

Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags.

Cancel

Previous

Create user

Step 1

Specify user details

Step 2

Set permissions

Step 3

Review and create

Retrieve password

Retrieve password

You can view and download the user's password below or email users instructions for signing in to the AWS Management Console. This is the only time you can view and download this password.

Console sign-in details

Email sign-in instructions

Console sign-in URL

<https://359416636630.signin.aws.amazon.com/console>

User name

Admin

Console password

XXXXXXXXXXXX Show

Cancel

Download .csv file

Return to users list

User created successfully

You can view and download the user's password and email instructions for signing in to the AWS Management Console.

View user

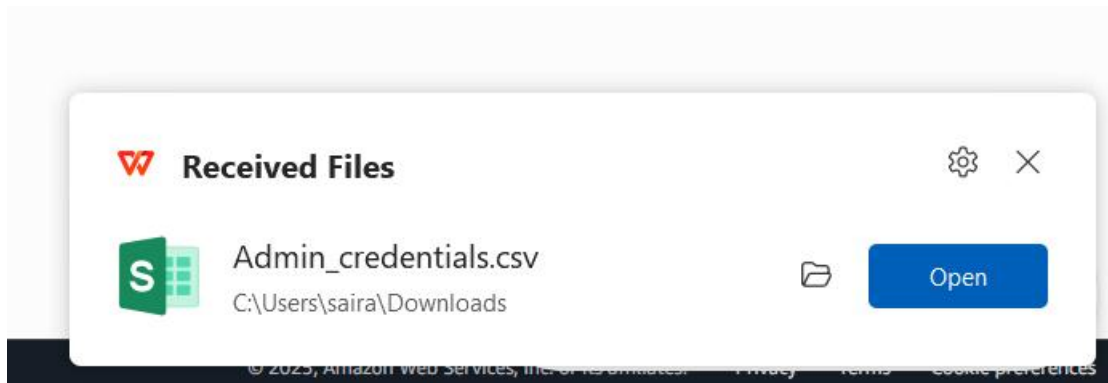
CloudShell

Feedback

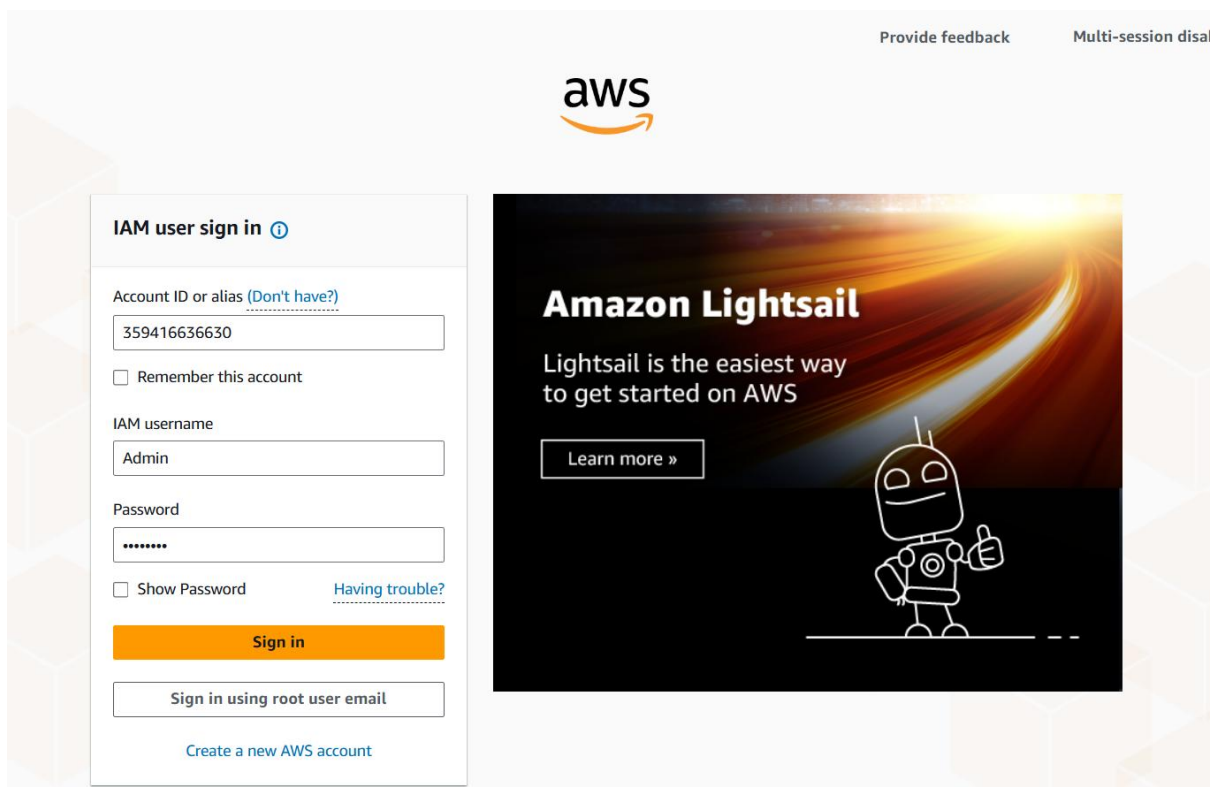
Console Mobile App

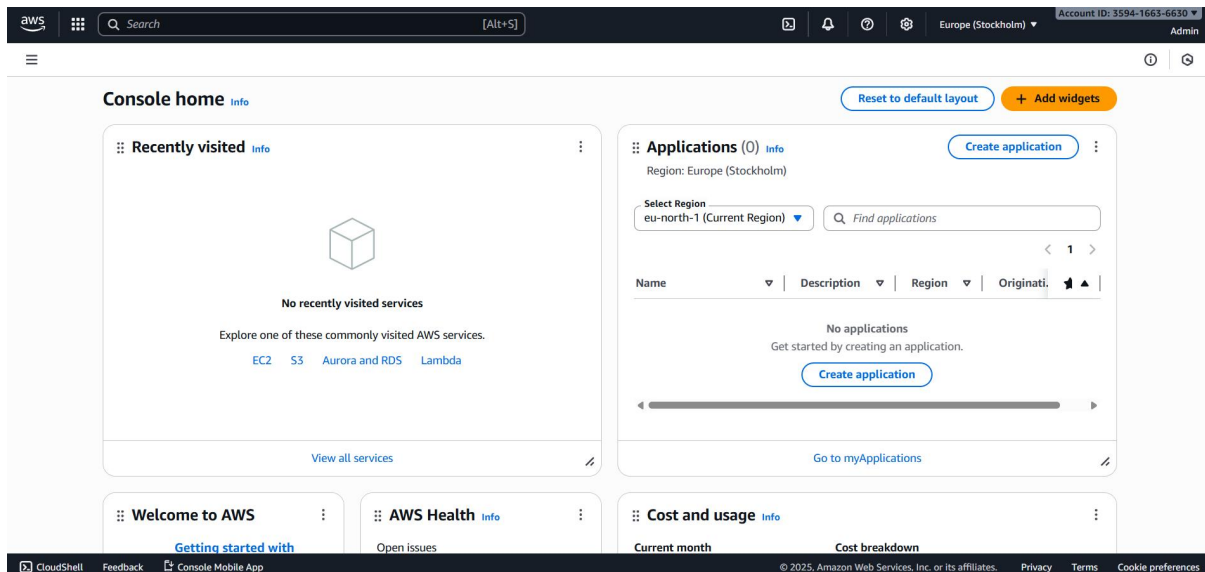
© 2025, Amazon Web Services, Inc. or its affiliates. [Privacy](#) [Terms](#) [Cookie preferences](#)

3. Download the Admin .csv and show its presence on your Windows host (do not display the password text):



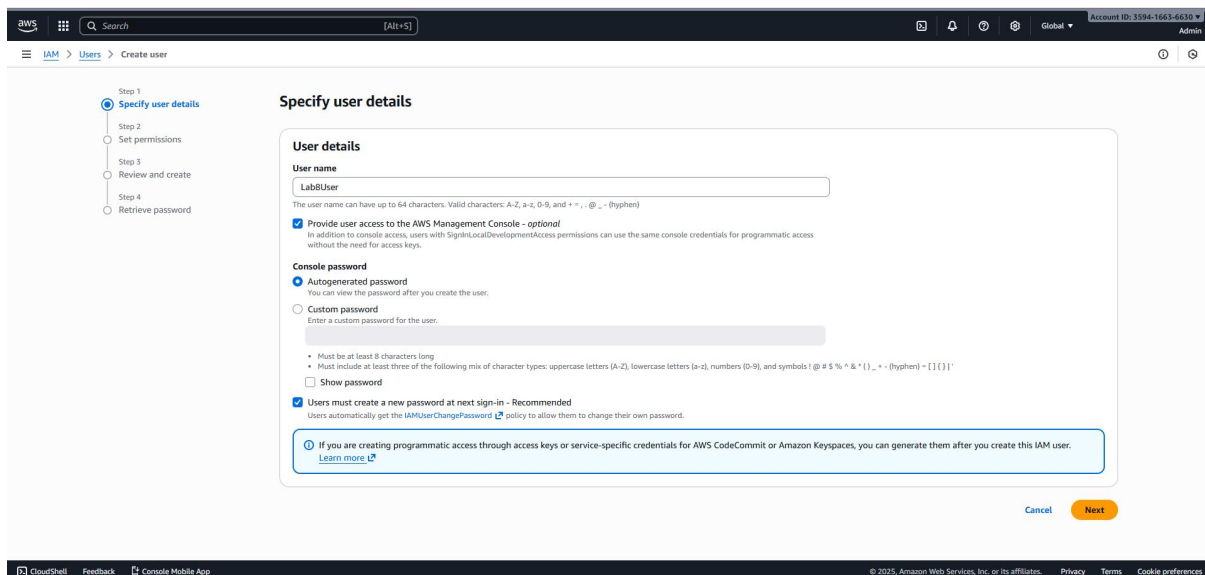
4. Sign out of root, then sign in using the Admin account (use the sign in URL from the .csv). Capture after successful Admin login:



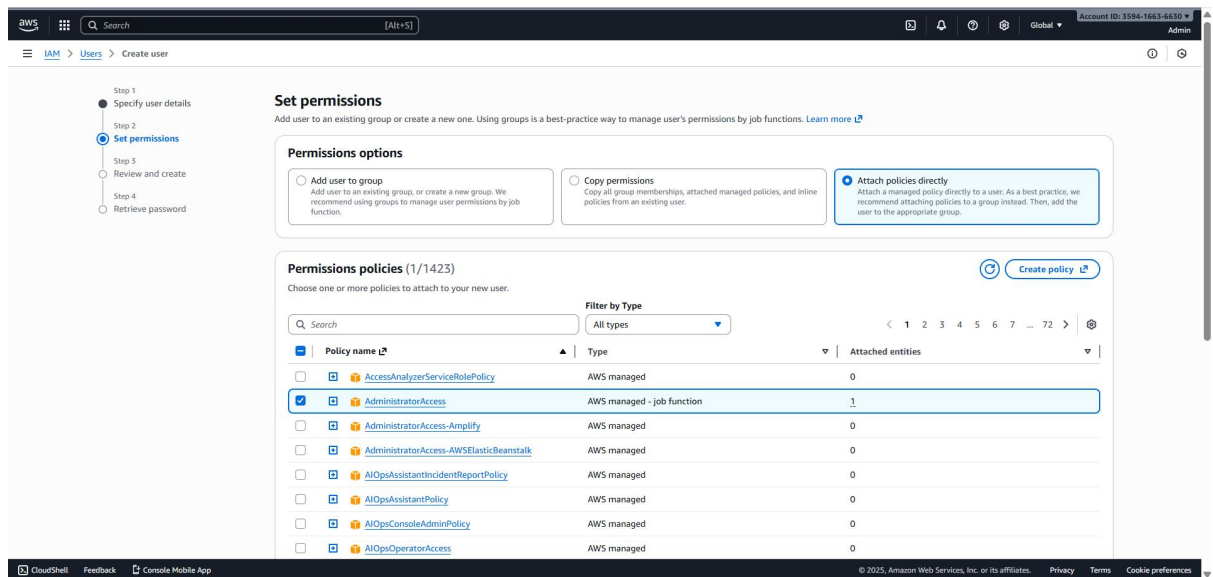


While logged in as Admin, create Lab8User:

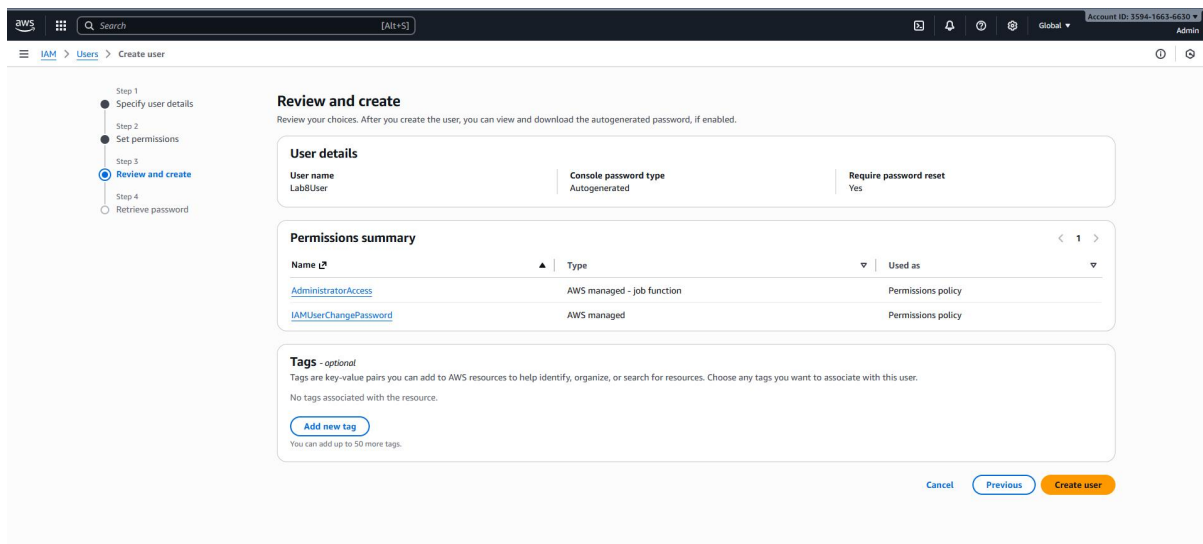
- IAM → Users → Create user
- Username: Lab8User
- Provide user access to the AWS Management Console



- Attach AdministratorAccess policy
- Capture the create-user success screen:



5. Download/save the Lab8User CSV on your Windows host (do not show password).



Received Files

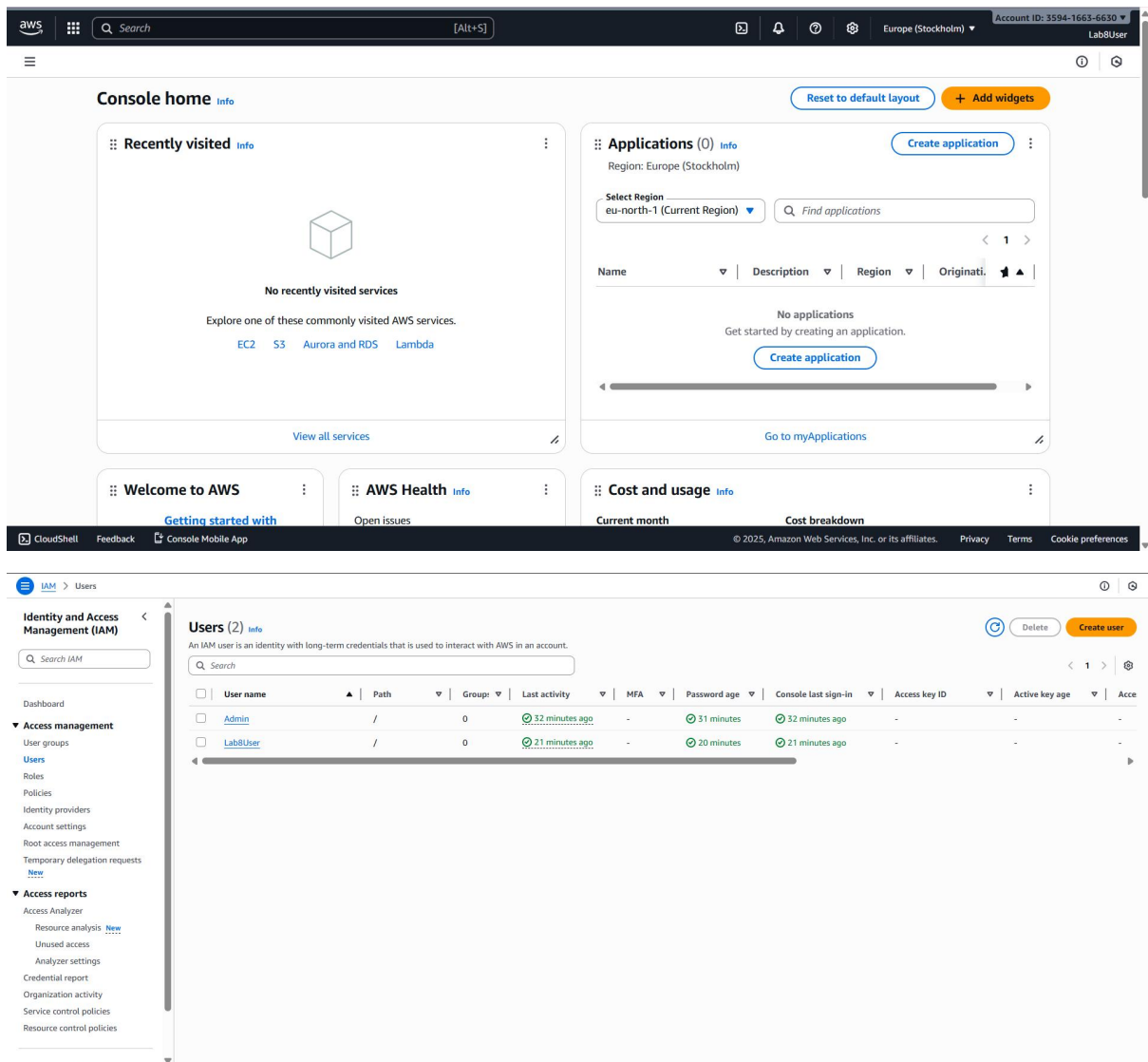


Lab8User_credentials.csv
C:\Users\saira\Downloads



Open

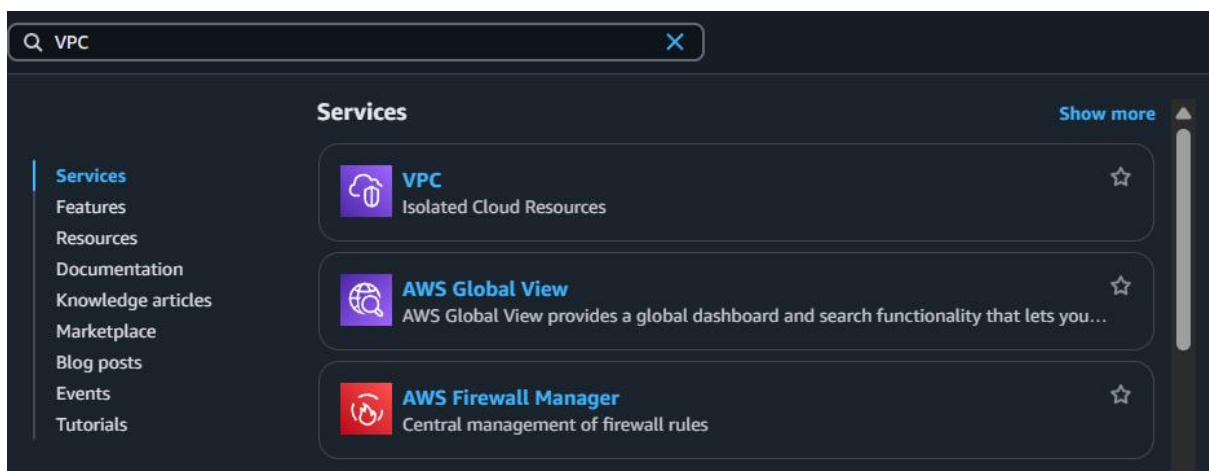
6. Logout Admin and login as Lab8User (use the Lab8User signin URL and credentials). Capture after login:

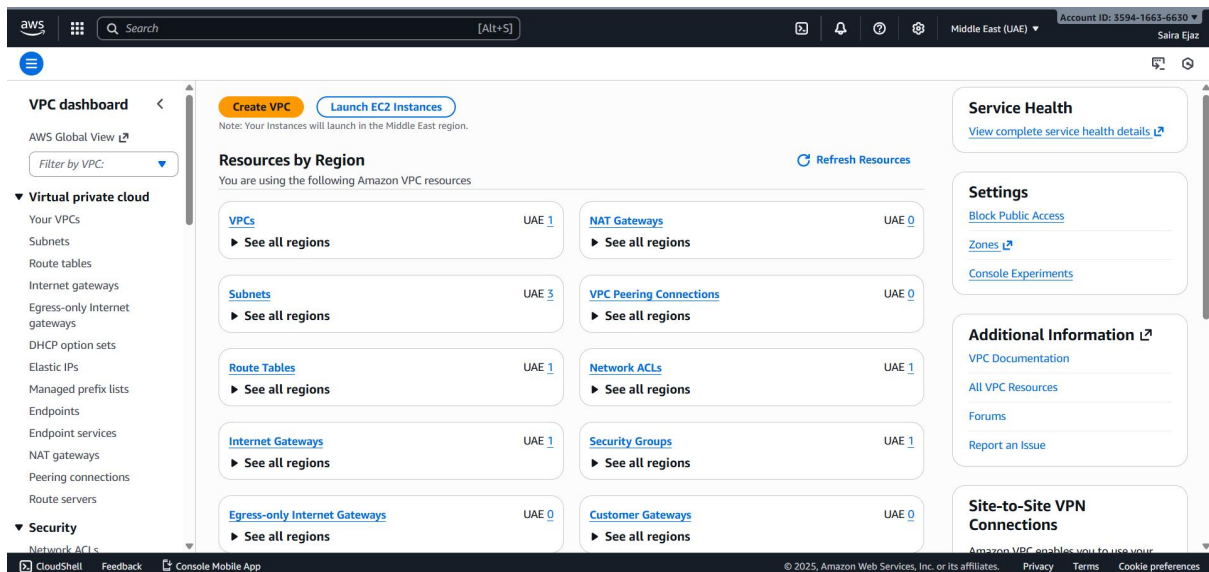


Task 3 — Inspect VPC resources (in UAE me-central-1)

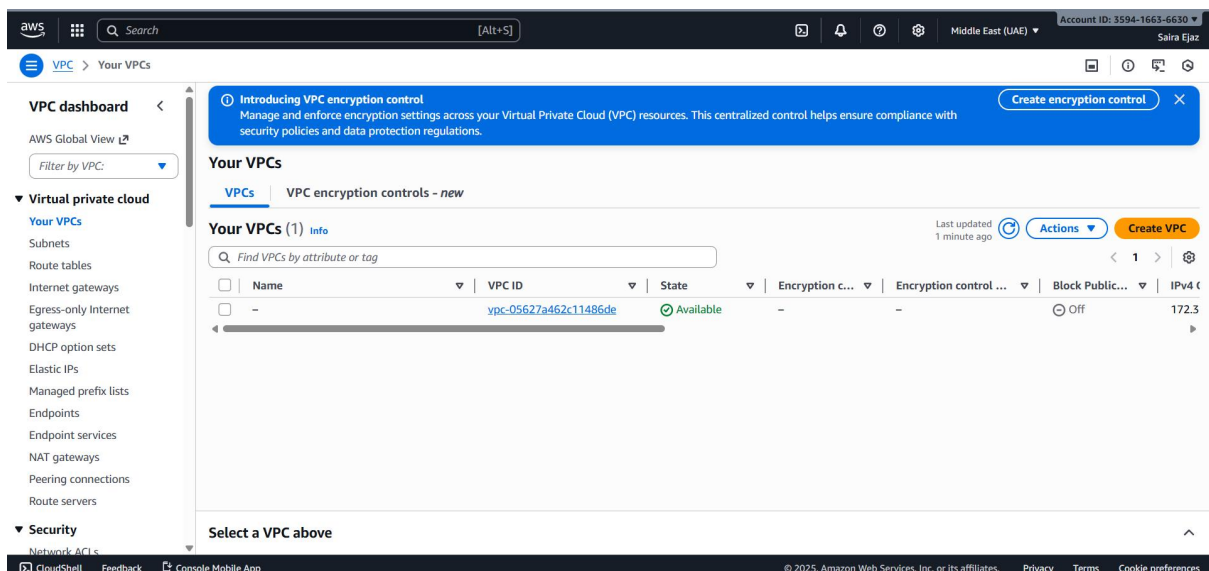
Goal: In the VPC console list the counts of specific resources.

1. Open VPC console (Alt+S → "VPC") while region is me-central-1.

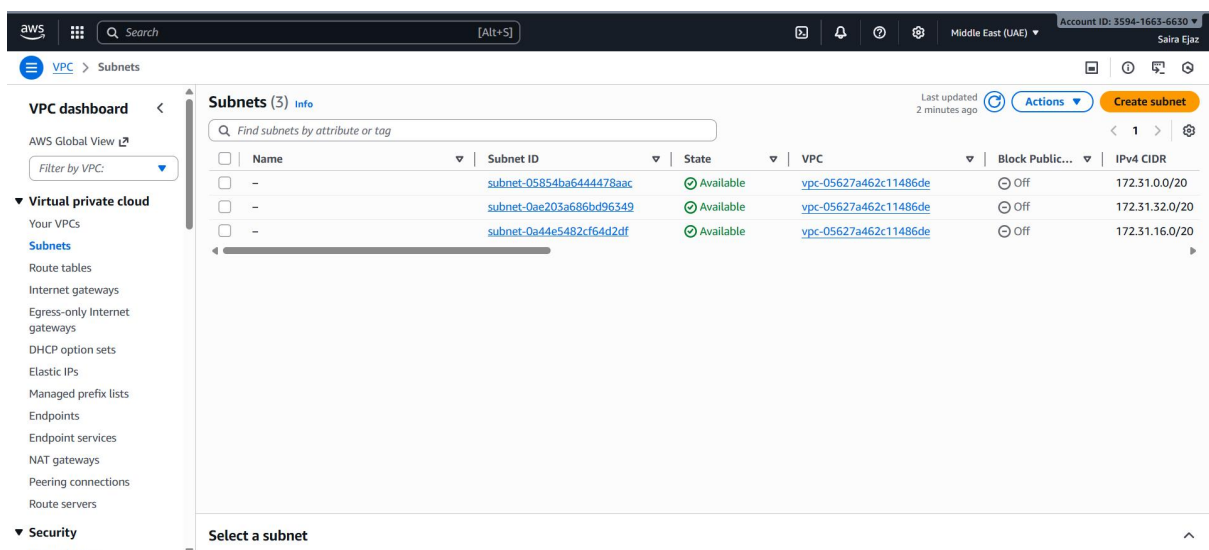




2. View VPCs list. Capture:



3. View Subnets list. Capture:



4. View Route Tables list. Capture:

The screenshot shows the AWS Management Console interface for the 'Route tables' section. The left sidebar contains the 'VPC dashboard' and a list of VPC resources. The main content area displays the 'Route tables (1)' list. A table with the following columns is shown: Name, Route table ID, Explicit subnet associ..., Edge associations, Main, and VPC. One route table is listed with ID 'rtb-04868885e737881c0' and VPC 'vpc-05627a462c11486de'. The interface includes a search bar, a 'Filter by VPC' dropdown, and a 'Create route table' button.

5. View Network ACLs list. Capture:

The screenshot shows the AWS Management Console interface for the 'Network ACLs' section. The left sidebar contains the 'Virtual private cloud' and 'Security' sections. The main content area displays the 'Network ACLs (1)' list. A table with the following columns is shown: Name, Network ACL ID, Associated with, Default, VPC ID, and Inbound. One network ACL is listed with ID 'acl-026692db7bed6597f' and VPC 'vpc-05627a462c11486de'. The interface includes a search bar, a 'Filter by VPC' dropdown, and a 'Create network ACL' button.

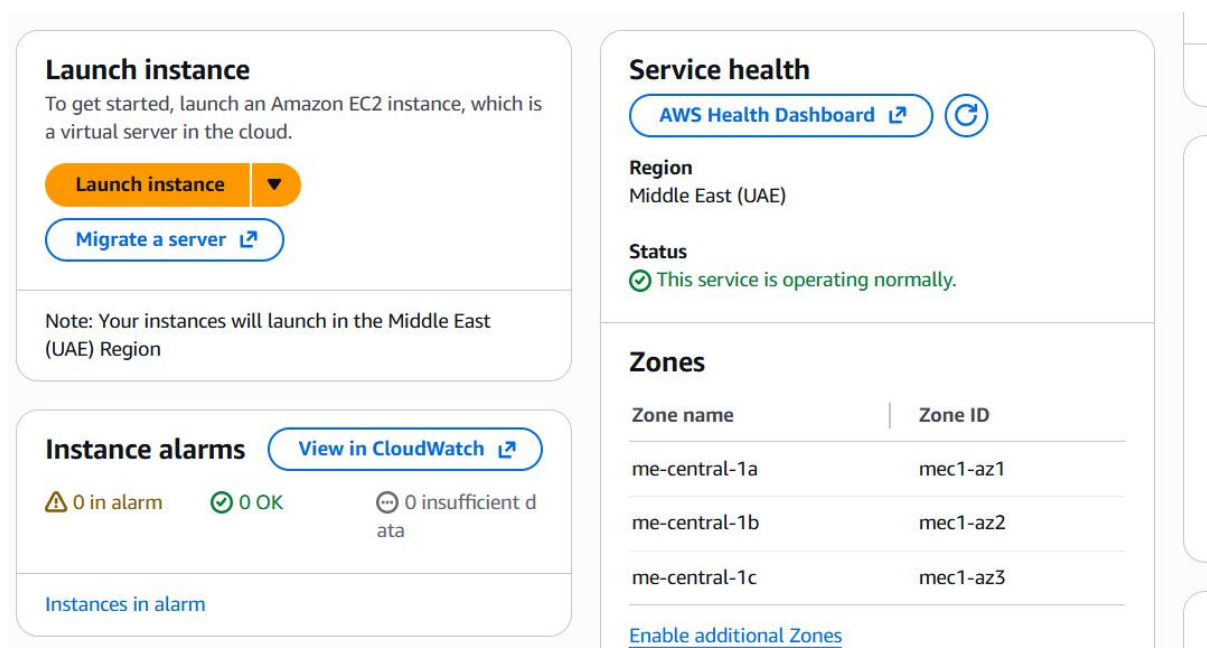
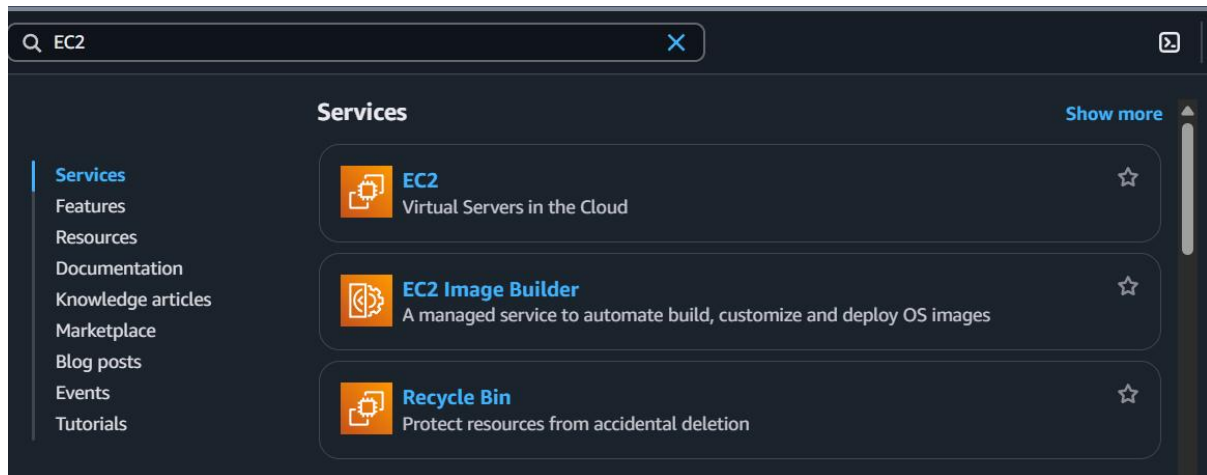
6. Task 3 summary (combine evidence):

The screenshot shows the AWS Management Console interface with the 'Recently visited' sidebar open. The sidebar lists various AWS services and categories, including 'VPC', 'IAM', 'Console Home', 'AWS Global View', and 'Billing and Cost Management'. The main content area displays the 'Network ACLs (1)' list, which is the same as in the previous screenshot. The interface includes a search bar, a 'Filter by VPC' dropdown, and a 'Create network ACL' button.

Task 4 — Launch EC2, SSH, install Docker & Docker Compose, deploy Gitea

Goal: Launch an EC2 instance named Lab8Machine, configure security group Lab8SecurityGroup allowing SSH from your IP, create an ED25519 key pair Lab8Key (.pem), SSH from Windows host, install Docker & Docker Compose, create compose.yaml from the repository, run docker compose up -d, allow inbound TCP port 3000, and open Gitea UI.

1. Open EC2 Console (Alt+S → "EC2") (me-central-1).



2. Instance Launch configuration (during review before launching). Configure:

- **Name: Lab8Machine**

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

Lab8Machine

[Add additional tags](#)

● AMI: Amazon Linux 2

 Search our full catalog including 1000s of application and OS images

Quick Start



[Browse more AMIs](#)

Including AMIs from
AWS, Marketplace and
the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI

ami-05e66df2bafcb7dea (64-bit (x86), uefi-preferred) / ami-0d90f860e9d3b1c73 (64-bit (Arm), uefi)

Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.9.20251117.1 x86_64 HVM kernel-6.1

Architecture

64-bit (x86) ▼

Boot mode

uefi-preferred

AMI ID

ami-05e66df2bafcb7dea

Publish Date

2025-11-17

Username [i](#)

ec2-user

[Verified provider](#)

● Instance type: t2.micro

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Ubuntu Pro base pricing: 0.016 USD per Hour

On-Demand Linux base pricing: 0.0125 USD per Hour On-Demand RHEL base pricing: 0.0413 USD per Hour

On-Demand SUSE base pricing: 0.0125 USD per Hour On-Demand Windows base pricing: 0.0217 USD per Hour

☒ All generations

[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

Select



[Create new key pair](#)

● Security group: Create Lab8SecurityGroup with SSH from My IP

▼ Network settings

Info

VPC - required

Info

vpc-05627a462c11486de

(default)

▼

172.31.0.0/16

↻

Subnet

Info

No preference

▼

↻

Create new subnet

Availability Zone

Info

No preference

▼

↻

Enable additional zones

Auto-assign public IP

Info

Enable

▼

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Security group name - required

Lab8SecurityGroup

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and ., -, /, #, @, !, +, =, &, ;, ' \$ *

Description - required

Info

launch-wizard-1 created 2025-11-28T16:19:34.621Z

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 203.81.236.245/32)

Remove

Type

Info

ssh

▼

Protocol

Info

TCP

Port range

Info

22

Source type

Info

My IP

▼

Name

Info

Q Add CIDR, prefix list or security group

203.81.236.245/32

×

Description - optional

Info

e.g. SSH for admin desktop

Add security group rule

- Storage: default

▼ Configure storage

Info

Advanced

1x

8

GiB

gp3

▼

Root volume, 3000 IOPS, Not encrypted

Add new volume

🕒 Click refresh to view backup information

↻

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems

Edit

- Key pair: Create Lab8Key (ED25519, .pem) and download the .pem file to your Windows host

Create a key pair or proceed without a key pair

i We noticed that you didn't select a key pair. If you want to be able to connect to your instance it is recommended that you create one or select an existing one.

☒ Create new key pair

☐ Proceed without key pair

Key pair name

Key pairs allow you to connect to your instance securely.

Enter key pair name

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☐ **RSA**
RSA encrypted private and public key pair

☒ **ED25519**
ED25519 encrypted private and public key pair

Private key file format

☒ **.pem**
For use with OpenSSH

☐ **.ppk**
For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. **You will need it later to connect to your instance.** [Learn](#)

Cancel

Launch instance

3. After launch, EC2 Instances list showing Lab8Machine in "running" state and public IPv4 visible.


```

[ec2-user@ip-172-31-13-62 ~]$ sudo yum install -y docker
Last metadata expiration check: 0:00:49 ago on Fri Nov 28 16:45:21 2025.
Dependencies resolved.

```

Package	Architecture	Version	Repository	Size
Installing:				
docker	x86_64	25.0.13-1.amzn2023.0.2	amazonlinux	46 M
Installing dependencies:				
container-selinux	noarch	4:2.242.0-1.amzn2023	amazonlinux	58 k
containerd	x86_64	2.1.4-1.amzn2023.0.2	amazonlinux	23 M
iptables-libs	x86_64	1.8.8-3.amzn2023.0.2	amazonlinux	401 k
iptables-nft	x86_64	1.8.8-3.amzn2023.0.2	amazonlinux	183 k
libgroup	x86_64	3.0-1.amzn2023.0.1	amazonlinux	75 k
libnetfilter_conntrack	x86_64	1.0.8-2.amzn2023.0.2	amazonlinux	58 k
libnetfilter_link	x86_64	1.0.1-19.amzn2023.0.2	amazonlinux	30 k
libnftnl	x86_64	1.2.2-2.amzn2023.0.2	amazonlinux	84 k
pipx	x86_64	2.5-1.amzn2023.0.3	amazonlinux	83 k
runc	x86_64	1.3.3-2.amzn2023.0.1	amazonlinux	3.9 M

```

Transaction Summary
Install 11 Packages

Total download size: 74 M
Installed size: 280 M
Downloading Packages:
(1/11): container-selinux-2.242.0-1.amzn2023.noarch.rpm                  1.0 MB/s | 58 kB  00:00
(2/11): iptables-libs-1.8.8-3.amzn2023.0.2.x86_64.rpm                 12 MB/s | 401 kB  00:00
(3/11): iptables-nft-1.8.8-3.amzn2023.0.2.x86_64.rpm                   6.3 MB/s | 183 kB  00:00
(4/11): libgroup-3.0-1.amzn2023.0.1.x86_64.rpm                         2.5 MB/s | 75 kB  00:00
(5/11): libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64.rpm        2.5 MB/s | 58 kB  00:00
(6/11): libnetfilter_link-1.0.1-19.amzn2023.0.2.x86_64.rpm            1.5 MB/s | 30 kB  00:00
(7/11): libnftnl-1.2.2-2.amzn2023.0.2.x86_64.rpm                      2.8 MB/s | 84 kB  00:00
(8/11): pipx-2.5-1.amzn2023.0.3.x86_64.rpm                            3.0 MB/s | 83 kB  00:00
(9/11): containerd-2.1.4-1.amzn2023.0.2.x86_64.rpm                    64 MB/s | 23 MB  00:00
(10/11): runc-1.3.3-2.amzn2023.0.1.x86_64.rpm                         28 MB/s | 3.9 MB  00:00
(11/11): docker-25.0.13-1.amzn2023.0.2.x86_64.rpm                     67 MB/s | 46 MB  00:00
Total
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction:
Preparing : runc-1.3.3-2.amzn2023.0.1.x86_64                          1/11
Installing : containerd-2.1.4-1.amzn2023.0.2.x86_64                  1/11
Running scriptlet: containerd-2.1.4-1.amzn2023.0.2.x86_64            2/11
Installing : pipx-2.5-1.amzn2023.0.3.x86_64                          3/11
Installing : libnftnl-1.2.2-2.amzn2023.0.2.x86_64                   4/11
Installing : libnetfilter_link-1.0.1-19.amzn2023.0.2.x86_64          5/11
Installing : iptables-libs-1.8.8-3.amzn2023.0.2.x86_64              6/11
Installing : iptables-nft-1.8.8-3.amzn2023.0.2.x86_64              7/11
Running scriptlet: iptables-nft-1.8.8-3.amzn2023.0.2.x86_64          8/11
Installing : libgroup-3.0-1.amzn2023.0.1.x86_64                      9/11
Running scriptlet: libgroup-3.0-1.amzn2023.0.1.x86_64

```

```

sudo mkdir -p /usr/local/lib/docker/cli-plugins
sudo curl -SL https://github.com/docker/docker/compose/releases/latest/download/docker-compose-linux-x86_64 -o /usr/local/lib/docker/cli-plugins/docker-compose
sudo chmod +x /usr/local/lib/docker/cli-plugins/docker-compose
sudo systemctl start docker

```

```

Complete!
[ec2-user@ip-172-31-13-62 ~]$ sudo mkdir -p /usr/local/lib/docker/cli-plugins
[ec2-user@ip-172-31-13-62 ~]$ sudo curl -SL https://github.com/docker/docker/compose/releases/latest/download/docker-compose-linux-x86_64 -o /usr/local/lib/docker/cli-plugins/docker-compose
% Total % Received % Xferd Average Speed Time Time Current
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
100 73.0M 100 73.0M 0 0 32.0M 0 0:00:02 0:00:02 --:--:-- 47.9M
[ec2-user@ip-172-31-13-62 ~]$ sudo chmod +x /usr/local/lib/docker/cli-plugins/docker-compose
[ec2-user@ip-172-31-13-62 ~]$ sudo systemctl start docker
[ec2-user@ip-172-31-13-62 ~]$

```

6. Create/edit compose.yaml on the EC2 instance (sudo vim compose.yaml) and paste content from the repo: [Gitea](https://github.com/docker/docker/compose/releases/latest/download/docker-compose-linux-x86_64) . While pasting, capture the editor content:

```

ec2-user@ip-172-31-13-62~
services:
  gitea:
    image: gitea/gitea:latest
    container_name: gitea
    environment:
      - DB_TYPE=postgres
      - DB_HOST=db:5432
      - DB_NAME=gitea
      - DB_USER=gitea
      - DB_PASSWD=gitea
    restart: always
    volumes:
      - gitea:/data
    ports:
      - 3000:3000
    extra_hosts:
      - "www.jenkins.com:host-gateway"
    networks:
      - webnet
  db:
    image: postgres:alpine
    container_name: gitea_db
    environment:
      - POSTGRES_USER=gitea
      - POSTGRES_PASSWORD=gitea
      - POSTGRES_DB=gitea
    restart: always
    volumes:
      - gitea_postgres:/var/lib/postgresql/data
    expose:
      - 5432
    networks:
      - webnet
volumes:
  gitea_postgres:
    name: gitea_postgres
  gitea:
    name: gitea
networks:
  webnet:
    name: webnet
# external: true

# Gitea is not allowed to webhook to Jenkins follow these steps
# 1) Go to Gitea Container
# 2) cat /data/gitea/conf/app.ini
# 3) echo "[webhook]" >> /data/gitea/conf/app.ini
# 4) echo "ALLOWED_HOST_LIST = 192.168.65.2" >> /data/gitea/conf/app.ini
# Gitea Tutorials : https://www.youtube.com/watch?v=dAM2CqH8TUA
~
~
~
-- INSERT --

```

7. Save and verify file exists:

```
[ec2-user@ip-172-31-13-62 ~]$ sudo vim compose.yaml
[ec2-user@ip-172-31-13-62 ~]$ ls -l compose.yaml
-rw-r--r--. 1 root root 1129 Nov 28 16:52 compose.yaml
[ec2-user@ip-172-31-13-62 ~]$
```

8. Add ec2-user to docker group, show groups before re-login, exit and reconnect, show groups after reconnect:

```
[ec2-user@ip-172-31-13-62 ~]$ groups
ec2-user adm wheel systemd-journal
[ec2-user@ip-172-31-13-62 ~]$ sudo usermod -aG docker $USER
[ec2-user@ip-172-31-13-62 ~]$ groups
ec2-user adm wheel systemd-journal
[ec2-user@ip-172-31-13-62 ~]$ exit
logout
Connection to 158.252.20.108 closed.
```

```
saira@DESKTOP-LOCUSEF MINGW64 ~ (main)
$ ssh -i "C:\Users\Saira\Downloads\Lab8Key.pem" ec2-user@158.252.20.108
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html

~_##### Amazon Linux 2023
~_ \#### \
~_ \### |
~_ \#/_--> https://aws.amazon.com/linux/amazon-linux-2023
~_ \V'_->
~_ \_./
~_ \_/
~_ /m/'_-->

Last login: Fri Nov 28 16:41:18 2025 from 203.81.236.245
[ec2-user@ip-172-31-13-62 ~]$ Groups
-bash: Groups: command not found
[ec2-user@ip-172-31-13-62 ~]$ groups
ec2-user adm wheel systemd-journal docker
[ec2-user@ip-172-31-13-62 ~]$
```

9. Run `docker compose up -d` from the directory with `compose.yaml`:

```
[lec2-user@ip-172-31-13-62 ~]$ docker compose up -d
[+] Running 17/17
  ✓ db Pulled
    ✓ 2d35ebdb57d9 Pull complete
    ✓ 87aaf2c1f39b Pull complete
    ✓ 84ae6d252b40 Pull complete
    ✓ 7dd90d8c5ae5 Pull complete
    ✓ d58703585b9c Pull complete
    ✓ 3174f2a40dc7 Pull complete
    ✓ 9e4eaf63327c Pull complete
    ✓ 4b67a2fbf223 Pull complete
    ✓ f6c5971200e2 Pull complete
    ✓ 9c312a93b89a Pull complete
  ✓ gitea Pulled
    ✓ 712adf69e56e Pull complete
    ✓ ceee20f5a0c8 Pull complete
    ✓ 510f786cf94b Pull complete
    ✓ edff09febfa9 Pull complete
    ✓ f84eddlc8d8b Pull complete
[+] Running 5/5
  ✓ Network webnet Created
  ✓ Volume gitea_postgres Created
  ✓ Volume gitea Created
  ✓ Container gitea_db Started
  ✓ Container gitea Started
[lec2-user@ip-172-31-13-62 ~]$
```

10. Edit the security group Lab8SecurityGroup inbound rules in the EC2 console: add Custom TCP rule port 3000 source 0.0.0.0/0 and save. Capture the inbound rules after saving:

The image shows two screenshots of the AWS Management Console. The top screenshot displays the 'Security Groups' page for the 'default VPC security group' (sg-0d8508f3b485c64b8). It lists two security groups: the default VPC security group and a launch-wizard-1 created group. The bottom screenshot shows the 'Edit inbound rules' page for the 'sg-0d8508f3b485c64b8 - default' security group. It shows two inbound rules: 'All traffic' and 'Custom TCP'. The 'Custom TCP' rule is selected, and its source is being edited to '0.0.0.0/0'. A warning message at the bottom states: 'Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.'

Security Groups (1/2)

	Description	Owner	Inbound rules count	Outbound rules count
7a462c11486de	default VPC security group	359416636630	1 Permission entry	1 Permission entry
7a462c11486de	launch-wizard-1 created 2025-11-28T1...	359416636630	1 Permission entry	1 Permission entry

sg-0d8508f3b485c64b8 - default

Inbound rules (1)

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
-	sgr-0e1a951eab153a4df	-	All traffic	All	All	sg-0d8508f3b485c64b8

Edit inbound rules

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules

Security group rule ID: sgr-0e1a951eab153a4df

Type	Protocol	Port range	Source	Description - optional
All traffic	All	All	Custom	
Custom TCP	TCP	3000	Anywhere...	

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Cancel **Preview changes** **Save rules**

11. From your Windows browser navigate to: <http://Public-IP:3000> — capture the Gitea setup/install page:

Initial Configuration

If you run Gitea inside Docker, please read the [documentation](#) before changing any settings.

Database Settings

Gitea requires MySQL, PostgreSQL, MSSQL, SQLite3 or TiDB (MySQL protocol).

Database Type *	PostgreSQL
Host *	db:5432
Username *	gitea
Password *	
Database Name *	gitea
SSL *	Disable
Schema	

Leave blank for database default ("public").

General Settings

Site Title *	Gitea: Git with a cup of tea
--------------	------------------------------

You can enter your company name here.

Repository Root Path *	/data/git/repositories
------------------------	------------------------

Server Domain *	158.252.20.108
-----------------	----------------

Domain or host address for the server.

SSH Server Port	22
-----------------	----

Port number your SSH server listens on. Leave empty to disable.

Gitea HTTP Listen Port *	3000
--------------------------	------

Port number the Gitea web server will listen on.

Gitea Base URL *	http://158.252.20.108:3000/
------------------	-----------------------------

Base address for HTTP(S) clone URLs and email notifications.

Log Path *	/data/gitea/log
------------	-----------------

Log files will be written to this directory.

☐ Enable Update Checker

Checks for new version releases periodically by connecting to gitea.io.

Optional Settings

► Email Settings

► Server and Third-Party Service Settings

► Administrator Account Settings

These configuration options will be written into: `/data/gitea/conf/app.ini` 

Install Gitea

Register

You are registering the first account in the system, which has administrator privileges. Please carefully remember your username and password. If you forget the username or password, please refer to the Gitea documentation to recover the account.

Username *

Email Address *
Password *
Confirm Password *

Register Account

or

 Sign in with OpenID

Already have an account? [Sign in now!](#)

IssuesPull RequestsMilestonesExplore

New Repository

A repository contains all project files, including revision history. Already hosting one elsewhere? [Migrate repository.](#)

Owner *  Admin

Some organizations may not show up in the dropdown due to a maximum repository count limit.

Repository Name *

Good repository names use short, memorable and unique keywords. A repository named "profile" or "profile-private" could be used to add a README.md for the user/organization profile.

Visibility ☐ Make repository private
Only the owner or the organization members if they have rights, will be able to see it.

Description

Template

Issue Labels

.gitignore

Choose which files not to track from a list of templates for common languages. Typical artifacts generated by each language's build tools are included on .gitignore by default.

12. Task 4 summary (combine evidence)

Instance summary for i-Oca35e9a2368eba7d (Lab8Machine) Info

Updated 16 minutes ago

Instance ID
i-Oca35e9a2368eba7d

IPv6 address
-

Public IPv4 address
158.252.20.108 | [open address](#)

Private IPv4 addresses
172.31.13.62

Instance state
Running

Public DNS
ec2-158-252-20-108-me-central-1.compute.amazonaws.com | [open address](#)

Hostname type
IP name: ip-172-31-13-62-me-central-1.compute.internal

Private IP DNS name (IPv4 only)
ip-172-31-13-62-me-central-1.compute.internal

Answer private resource DNS name
IPv4 (A)

Auto-assigned IP address
158.252.20.108 [Public IP]

Instance type
t3.micro

VPC ID
vpc-05627a462c11486de

Subnet ID
subnet-05854ba644478aac

IAM Role
-

Instance ARN
arn:aws:ec2:me-central-1:359416636630:instance/i-Oca35e9a2368eba7d

IMDSv2
Required

Elastic IP addresses
-

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | [Learn more](#)

Auto Scaling Group name
-

Managed
false

Operator
-

Security Groups (1/2) Info

Find security groups by attribute or tag

Name	Security group ID	Security group name	VPC ID	Description
-	sg-048508f3b485c64b8	default	vpc-05627a462c11486de	default VPC security group
-	sg-04619d3a63e52934e	Lab8SecurityGroup	vpc-05627a462c11486de	launch-wizard-1 created 2025-11-28T1...

sg-04619d3a63e52934e - Lab8SecurityGroup

Inbound rules (2)

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
-	sgr-0a0bcb47abcc65255	IPv4	Custom TCP	TCP	3000	0.0.0.0/0
-	sgr-099859b98904f56b4	IPv4	SSH	TCP	22	203.81.236.245/32

Admin / Lab8

Code Issues Packages Projects Wiki

Quick Guide

Clone this repository Need help cloning? Visit [Help](#).

New File Upload File HTTP SSH <http://158.252.20.108:3000/Admin/Lab8.git>

Creating a new repository on the command line

```
touch README.md
git init
git checkout -b main
git add README.md
git commit -m "first commit"
git remote add origin http://158.252.20.108:3000/Admin/Lab8.git
git push -u origin main
```

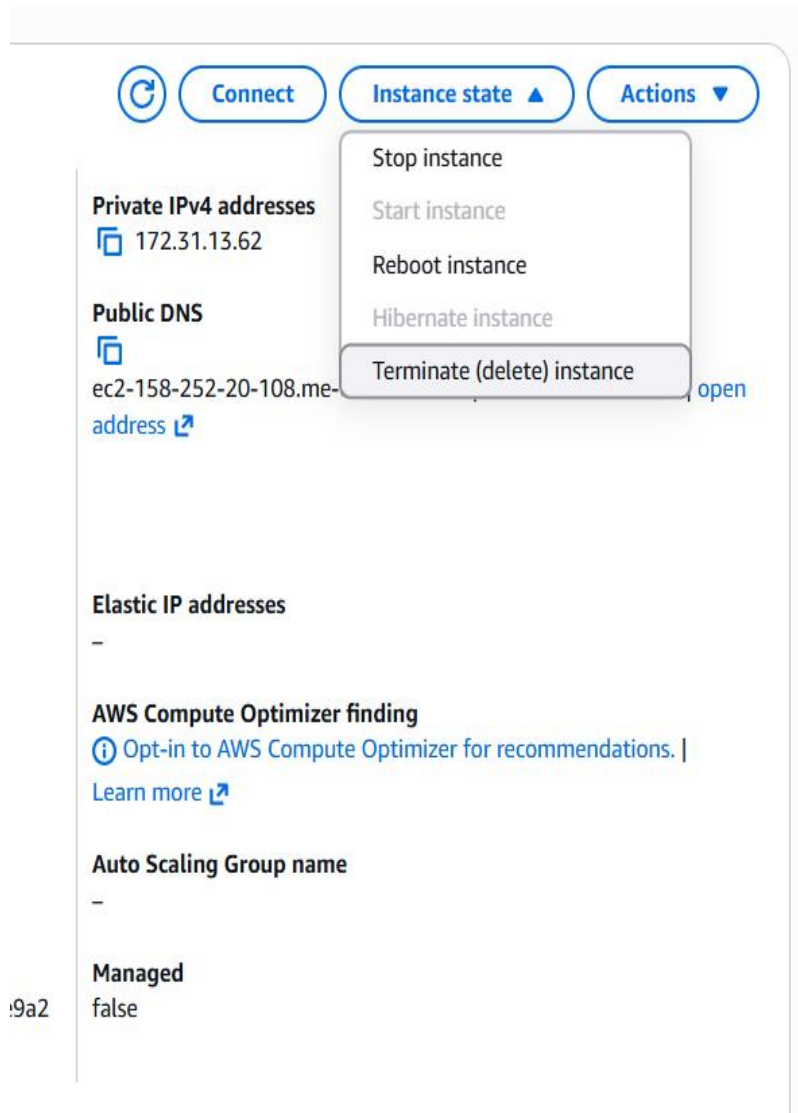
Pushing an existing repository from the command line

```
git remote add origin http://158.252.20.108:3000/Admin/Lab8.git
git push -u origin main
```

Cleanup — Remove resources to avoid charges

Cleanup steps and required screenshots:

1. Terminate the EC2 instance Lab8Machine



Terminate (delete) instance



On an EBS-backed instance, the default action is for the root EBS volume to be deleted when the instance is terminated. Storage on any local drives will be lost.

Are you sure you want to terminate these instances?

Instance ID	Termination protection
i-0ca35e9a2368eba7d (Lab8Machine)	Disabled

To confirm that you want to delete the instances, choose the terminate button below. Instances with termination protection enabled will not be terminated. Terminating the instance cannot be undone.

Skip OS shutdown

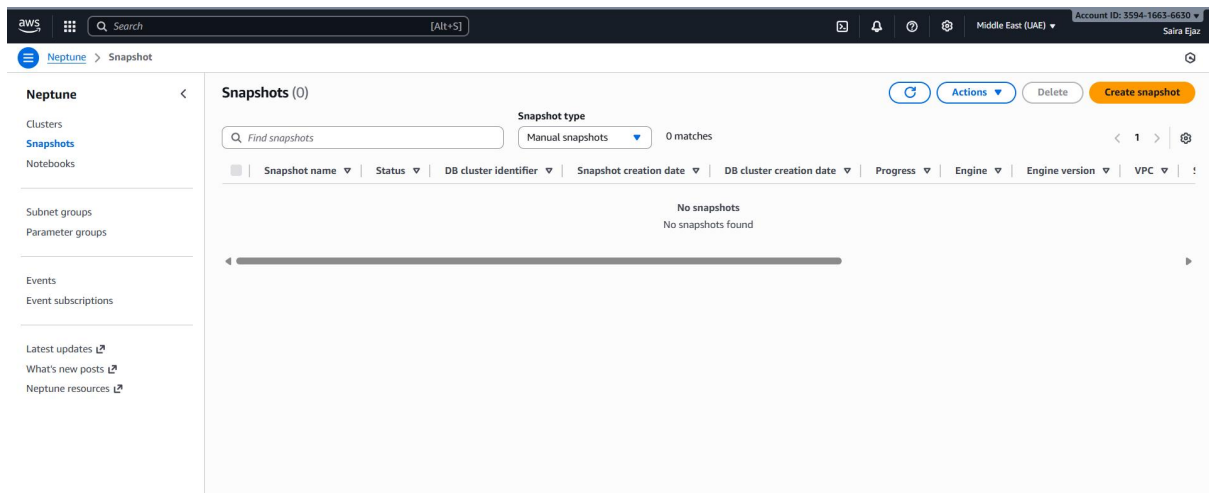
This option skips the graceful OS shutdown process. Use only when your instance must be stopped immediately, such as during an emergency or failover.

☐ Skip OS shutdown

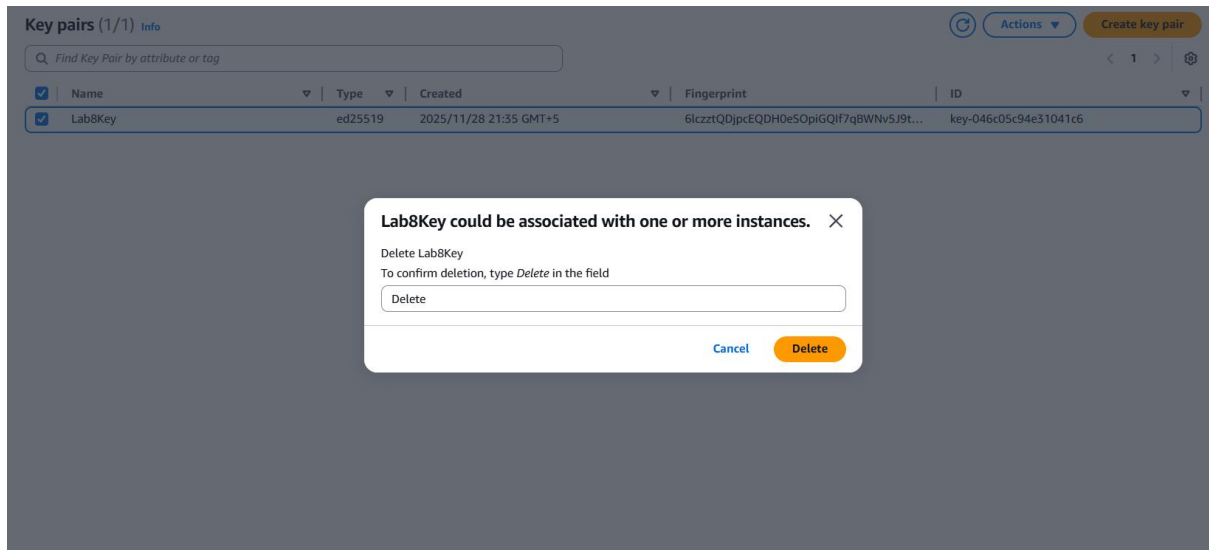
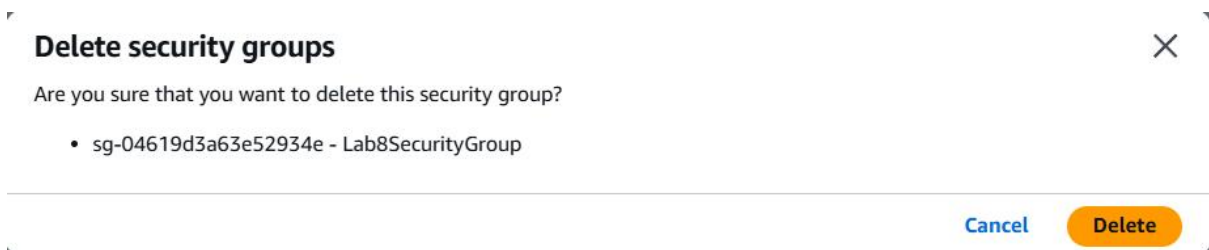
Cancel

Terminate (delete)

2. Delete associated EBS volumes and snapshots (if any).



3. Delete security group Lab8SecurityGroup and key pair Lab8Key from the EC2 console (after instances terminated).



4. Delete IAM users Lab8User and any access keys.

Delete Lab8User?



Delete **Lab8User** permanently? This will also delete all its user data, security credentials and inline policies.

User name	Last activity
Lab8User	4 hours ago

Note: Recent activity usually appears within 4 hours. Data is stored for a maximum of 365 days, depending when your region began supporting this feature. [Learn more](#)

To avoid accidental deletions, we ask you to provide additional written consent.

To confirm this deletion, type "confirm".

Cancel

Delete user

5. Final cleanup summary (show billing or resource groups with no active resources if possible).

